

TRIANGLE

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CLASS - 9th

Q1 In quadrilateral ABCD $AC=AD$ and AB bisects $\angle A$. Show that $\triangle ABC \cong \triangle ABD$. What can you say about BC and BD?

Sol: $\rightarrow$ In $\triangle ABC$ and $\triangle ABD$, we have

$$AC = AD \quad [\text{Given}]$$

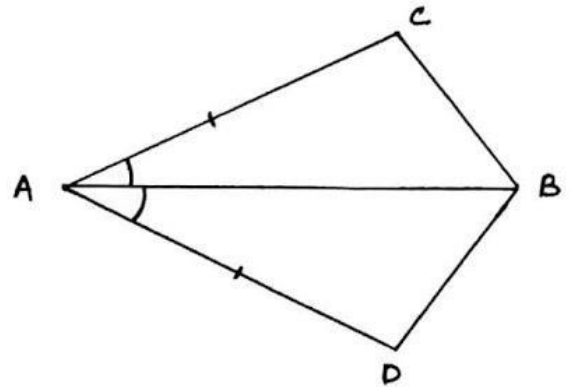
$$\angle CAB = \angle DAB$$

$$AB = AB$$

$$\therefore \triangle ABC \cong \triangle ABD$$

[By SAS congruence]

Therefore $BC = BD$ (CPCT) Ans.



Q2 ABCD is a quadrilateral in which $AD=BC$ and $\angle DAB = \angle CBA$. Prove that

i) $\triangle ABD \cong \triangle BAC$

ii) $BD = AC$

iii) $\angle ABD = \angle BAC$

Sol: $\rightarrow$ In the given figure ABCD is a quadrilateral in which $AD=BC$ and $\angle DAB = \angle CBA$

In $\triangle ABD$ and $\triangle BAC$, we have

$$AD = BC \quad [\text{Given}]$$

$$\angle DAB = \angle CBA \quad "$$

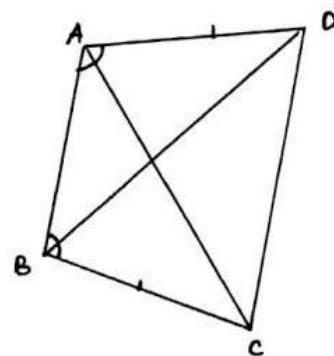
$$AB = AB \quad [\text{Common}]$$

$$\therefore \triangle ABD \cong \triangle BAC \quad [\text{By SAS}]$$

$$\therefore BD = AC \quad [\text{CPCT}]$$

$$\text{and } \angle ABD = \angle BAC \quad [\text{CPCT}]$$

Proved



Q3 AD and BC are equal perpendiculars to a line segment AB. Show that CD bisects AB.

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Sol: $\rightarrow$ In $\triangle AOD$ and $\triangle BOC$ we have

$$\angle AOD = \angle BOC$$

[Vertically opposite angles]

$$\angle CBO = \angle DAO \text{ [Each} = 90^\circ \text{]}$$

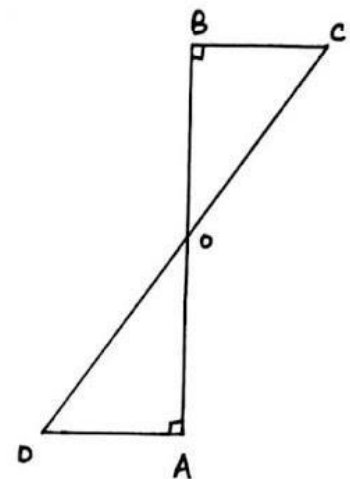
$$\text{and } AD = BC \text{ [Given]}$$

$$\therefore \triangle AOD \cong \triangle BOC \text{ [By AAS Congruence]}$$

$$\text{Also } AO = BO \text{ [CPCT]}$$

Hence, CD bisects AB

Proved.



Q4 l and m are two parallel lines intersected by another pair of parallel lines p and q. Show that $\triangle ABC \cong \triangle CDA$

Sol. In the given figure, ABCD is a parallelogram in which AC is a diagonal i.e. $AB \parallel DC$ and $BC \parallel AD$

In $\triangle ABC$ and $\triangle CDA$ we have

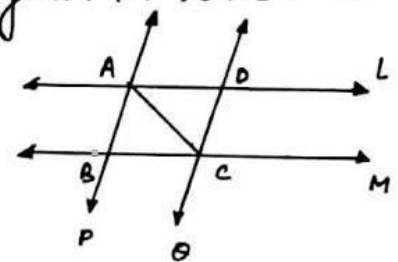
$$\angle BAC = \angle DCA \text{ [Alternate angle]}$$

$$\angle BCA = \angle DAC \text{ "}$$

$$AC = AC \text{ [common]}$$

$$\therefore \triangle ABC \cong \triangle CDA \text{ [By ASA congruence]}$$

Proved



Q5 Line l is the bisector of an angle A and B is any point on l, BP and BQ are perpendicular from B to the arms of $\angle A$ (see fig) Show that

$$i) \triangle APB \cong \triangle AQB$$

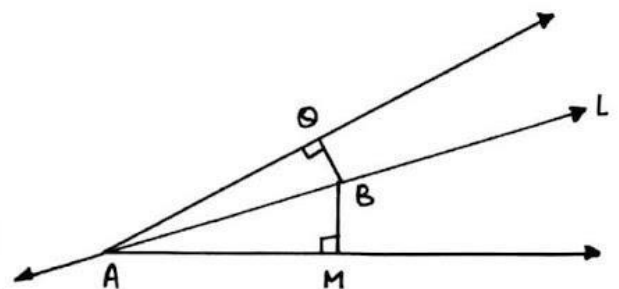
$$ii) BP = BQ \text{ or B is}$$

equidistant from the arms of $\angle A$

Sol: $\rightarrow$ In $\triangle APB$ and $\triangle AQB$, we have

$$\angle PAB = \angle QAB$$

[l is the bisector of $\angle A$]



$$\angle APB = \angle AOB \quad [\text{Each } 90^\circ]$$

$$AB = AB \quad [\text{Common}]$$

$$\therefore \triangle APB \cong \triangle AOB \quad [\text{By AAS Congruence}]$$

$$\text{Also } BP = BO \quad \text{By CPCT}$$

ie B is equidistant from the arms of LA

Hence Proved

Q6 In the figure, $AC = AE$, $AB = AD$ and $\angle BAD = \angle EAC$. Show that $BC = DE$.

$$\text{Sol:} \rightarrow \angle BAD = \angle EAC \quad [\text{Given}]$$

$$\Rightarrow \angle BAD + \angle DAC = \angle EAC + \angle DAC$$

[Adding $\angle DAC$ to both sides]

$$\Rightarrow \angle BAC = \angle EAC \quad - (i)$$

Now in $\triangle ABC$ and $\triangle ADE$, we have

$$AB = AD \quad [\text{Given}]$$

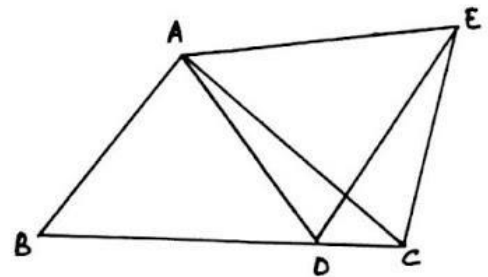
$$AC = AE \quad [\text{Given}]$$

$$\Rightarrow \angle BAC = \angle DAE \quad [\text{From i}]$$

$$\therefore \triangle ABC \cong \triangle ADE \quad [\text{By SAS congruence}]$$

$$\Rightarrow BC = DE \quad [\text{By CPCT}]$$

Hence proved.



Q7 AB is a line segment and P is its mid-point D and E are points on the same side of AB such that $\angle BAD = \angle ABE$ and $\angle EPA = \angle DPB$ [See fig] Show that

$$i) \triangle DAP \cong \triangle EBP$$

$$ii) AD = BE$$

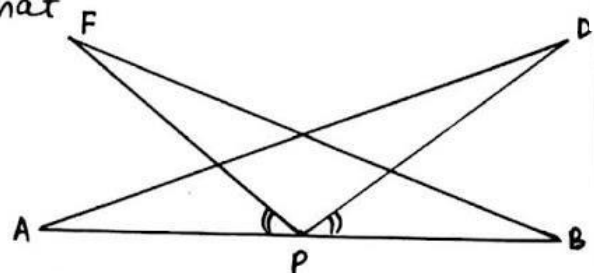
Sol: $\rightarrow$ In $\triangle DAP$ and $\triangle EBP$, we have

$$AP = BP \quad [P \text{ is mid-point of}]$$

[Line segment AB]

$$\angle BAD = \angle ABE \quad [\text{Given}]$$

$$\angle EPB = \angle DPA$$



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$$\therefore \triangle DPA \cong \triangle EPB \quad [ASA]''$$

$$\Rightarrow AD = BE \quad [By CPCT]$$

Hence proved

Q8 In right triangle ABC, right angled at C, M is the mid-point of hypotenuse AB. C is joined to M and produced to a point D such that $DM = CM$. Point D is joined to point B

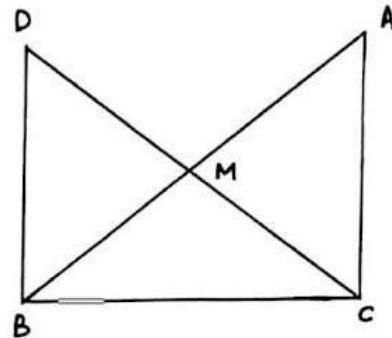
Show that:

$$i) \triangle AMC \cong \triangle BMD$$

$$ii) \angle OBC \text{ is a right angle.}$$

$$iii) \triangle OBC \cong \triangle ACB$$

$$iv) CM = \frac{1}{2} AB$$



Sol: $\rightarrow$ In $\triangle BMD$ and $\triangle AMC$

$$DM = CM \quad [Given]$$

$$BM = AM \quad [M \text{ is the mid-point of } AB]$$

$$\angle DMD = \angle AMC \quad [V.O.A]$$

$$\therefore \triangle AMC \cong \triangle BMD \quad [By SAS]$$

Hence Proved.

$$ii) AC \parallel BD \quad [\angle DBM \text{ and } \angle CAM \text{ are alternate angle}]$$

$$\Rightarrow \angle OBC + \angle ACB = 180^\circ \quad [Sum \text{ of co-interior angles}]$$

$$\angle OBC + \angle ACB = 180 \quad [\angle ACB = 90^\circ]$$

$$\therefore \angle OBC = 90^\circ$$

Hence proved.

iii) In $\triangle OBC$ and $\triangle ACB$, we have

$$OB = AC \quad [CPCT]$$

$$BC = BC \quad [common]$$

$$\angle OBC = \angle ACB \quad [Each 90^\circ]$$

$$\therefore \triangle OBC \cong \triangle ACB \quad [By SAS] \text{ Proved}$$

iv) As $\triangle OBC \cong \triangle OCB$

$\therefore AB = CD$ [By CPCT]

$$\Rightarrow \frac{1}{2}AB = \frac{1}{2}CD$$

$$\text{Hence, } \frac{1}{2}AB = CM \quad [CM = \frac{1}{2}CD]$$

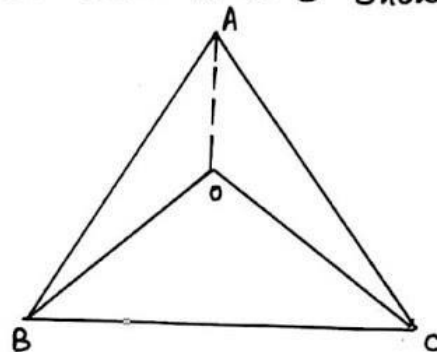
Hence proved.

Exercise 7.2

Q1 In an isosceles triangle ABC , with $AB = AC$, the bisector of $\angle B$ and $\angle C$ intersect each other at O . Join A to O show that

i) $OB = OC$

ii) AO bisect $\angle A$



Sol. (i) $AB = AC$ (Given)

$$\Rightarrow \angle ABC = \angle ACB$$

$$\frac{1}{2} \angle ABC = \frac{1}{2} \angle ACB$$

$$\Rightarrow \angle CBO = \angle BCO \quad [OB \text{ and } OC \text{ are bisector of } \angle B \text{ and } \angle C]$$

$$\Rightarrow OB = OC \quad [\text{Sides opposite to equal angles are equal}]$$

$$\text{Again } \frac{1}{2} \angle ABC = \frac{1}{2} \angle ACB$$

$$\Rightarrow \angle ABO = \angle ACO \quad [\because OB \text{ and } OC \text{ are bisector of } \angle B \text{ and } \angle C \text{ respectively}]$$

In $\triangle ABO$ and $\triangle ACO$, we have

$$AB = AC \quad [\text{Given}]$$

$$OB = OC \quad [\text{Proved above}]$$

$$\angle ABO = \angle ACO \quad ["]$$

$$\therefore \triangle ABO \cong \triangle ACO \quad [\text{SAS congruence}]$$

$$\Rightarrow \angle BAO = \angle CAO$$

$\Rightarrow AO$ bisect $\angle A$ Hence proved.

Q2 In $\triangle ABC$, AD is the perpendicular bisector of BC. (see figure). Show that $\triangle ABC$ is an isosceles triangle in which $AB = AC$.

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Sol: $\rightarrow$ In $\triangle ABD$ and $\triangle ACD$, we have

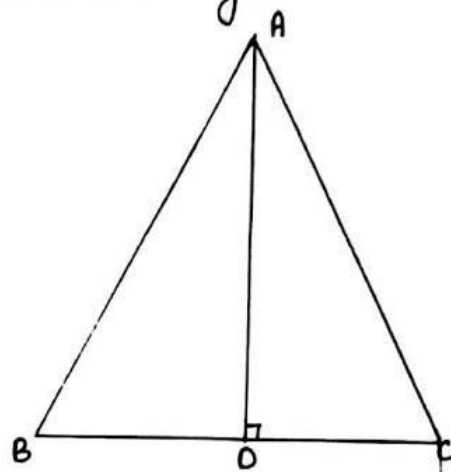
$$\angle ADB = \angle ADC \quad [\text{Each } 90^\circ]$$

$$BD = CD \quad [AD \text{ bisect } BC]$$

$$AD = AD \quad [\text{Common}]$$

$$\therefore \triangle ABD \cong \triangle ACD \quad [\text{SAS}]$$

$$\therefore AB = AC \quad [\text{By CPCT}]$$



Hence $\triangle ABC$ is an isosceles triangle Proved.

Q3: $\rightarrow$ ABC is an isosceles triangle in which altitude BE and CF are drawn to equal sides AC and AB respectively. Show that these altitudes are equal.

Sol: $\rightarrow$ In $\triangle ABC$

$$AB = AC \quad [\text{Given}]$$

$$\Rightarrow \angle B = \angle C \quad [\text{Angles opp to equal sides of triangle are equal}]$$

Now, in right triangles BFC and CEB

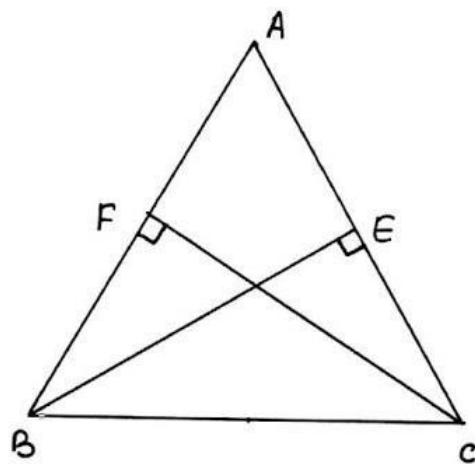
$$\angle BFC = \angle CEB \quad [\text{Each } 90^\circ]$$

$$\angle FBC = \angle ECB \quad [\text{Proved above}]$$

$$BC = BC \quad [\text{Common}]$$

$$\therefore \triangle BFC \cong \triangle CEB \quad [\text{AAS}]$$

$$\text{Hence, } BE = CF \quad [\text{CPCT}]$$



Q4 ABC is a triangle in which altitude BE and CF to sides AC and AB are equal show that

$$1) \triangle ABE \cong \triangle ACF$$

$$2) AB = AC$$

Sol: → (i) In $\triangle ABE$ and $\triangle ACF$ we have

$$BE = CF \text{ [Given]}$$

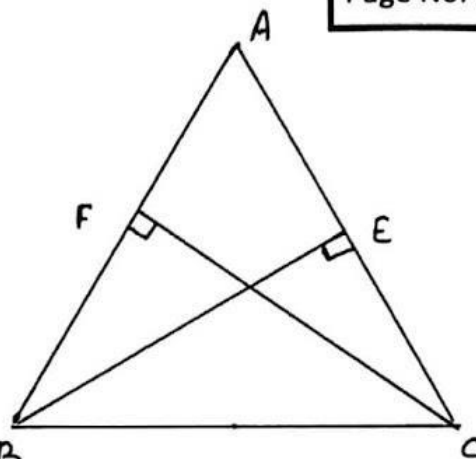
$$\angle BAE = \angle CAF \text{ [Common]}$$

$$\angle BEA = \angle CFA \text{ [Each } 90^\circ]$$

$$\text{So } \triangle ABE \cong \triangle ACF \text{ [AAS]}$$

(ii) Also $AB = AC$ [By CPCT]

ie $\triangle ABC$ is an isosceles triangle.



Q5 $\triangle ABC$ and $\triangle DCB$ are two triangles on the same base BC . Show that $\angle ABD = \angle ACD$

Sol: → In isosceles $\triangle ABC$, we have

$$AB = AC$$

$$\therefore \angle ABC = \angle ACB \text{ — (i)}$$

[Angles opp to equal sides are equal]

Now, in isosceles $\triangle DCB$, we have

$$BD = CD$$

$$\angle DBC = \angle DCB \text{ — (ii)}$$

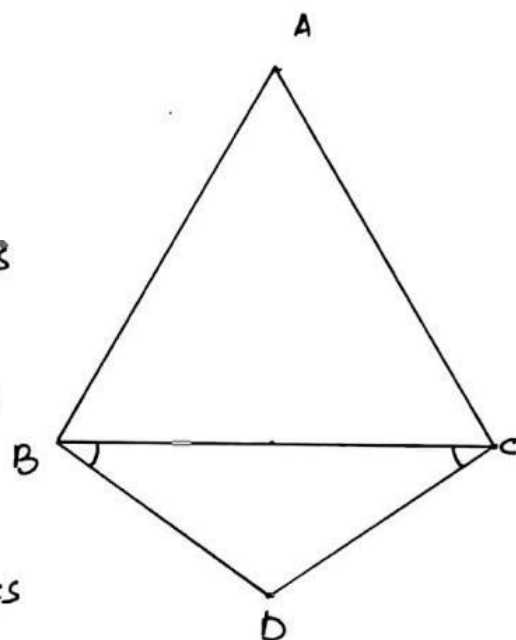
[Angles opp to equal sides are equal]

Adding (i) and (ii) we have

$$\angle ABC + \angle DBC = \angle ACB + \angle DCB$$

$$\Rightarrow \angle ABD = \angle ACD$$

Proved.



Q6: → $\triangle ABC$ is an isosceles triangle in which $AB = AC$. Side BA is produced to D such that $AD = AB$. Show that $\angle BCD$ is a right angle.

Sol: → $AB = AC$ [Given]

$$\angle ACB = \angle ABC \text{ — (i)}$$

[Angles opp to equal sides are equal]

$$AB = AD \text{ [Given]}$$

$$\therefore AD = AC \text{ [}\because AB = AC\text{]}$$

$$\therefore \angle ACD = \angle ADC \text{ — (ii)}$$

[Angles opp. to equal sides are also equal]

Adding (i) and (ii)

$$\angle ACB + \angle ACD = \angle ABC + \angle ADC$$

$$\Rightarrow \angle BCD = \angle ABC + \angle ADC \text{ — (iii)}$$

Now in $\triangle BCD$, we have

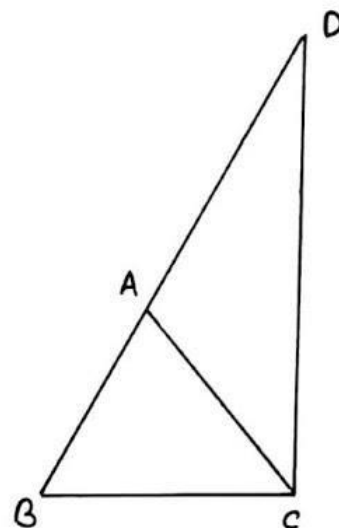
$$\angle BCD + \angle DBC + \angle BDC = 180^\circ \text{ [Angle sum property of a triangle]}$$

$$\therefore \angle BCD + \angle BCD = 180^\circ$$

$$2\angle BCD = 180^\circ$$

$$\therefore \angle BCD = 90^\circ$$

Hence $\angle BCD = 90^\circ$ or a right angle



Q7 $\triangle ABC$ is a right angled triangle in which $\angle A = 90^\circ$ and $AB = AC$. Find $\angle B$ and $\angle C$.

Sol $\rightarrow$ In $\triangle ABC$, we have

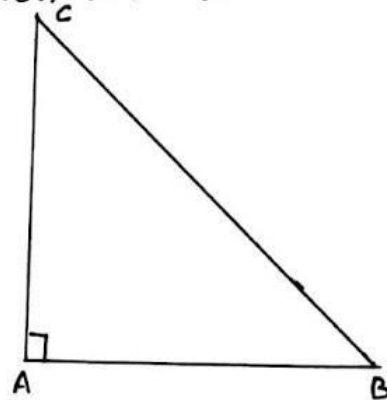
$$\left. \begin{array}{l} \angle A = 90^\circ \\ \text{and } AB = AC \end{array} \right\} \text{ [Given]}$$

We know that angles opposite to equal sides of an isosceles triangle are equal.

$$\text{So } \angle B = \angle C$$

Since $\angle A = 90^\circ$ therefore sum of remaining two angles $= 90^\circ$

$$\therefore \angle B = \angle C = 45^\circ$$



Q8 Show that the angles of an equilateral triangle are 60° each

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Sol: $\rightarrow$ As $\triangle ABC$ is an equilateral triangle

$$\text{So } AB = BC = AC$$

$$\text{Now } AB = AC$$

$$\therefore \angle ACB = \angle ABC \text{ --- (i)}$$

[Angles opp to equal sides are also equal]

$$\text{Also } BC = AC$$

$$\Rightarrow \angle BAC = \angle ABC \text{ --- (ii) [same reason]}$$

Now in $\triangle ABC$,

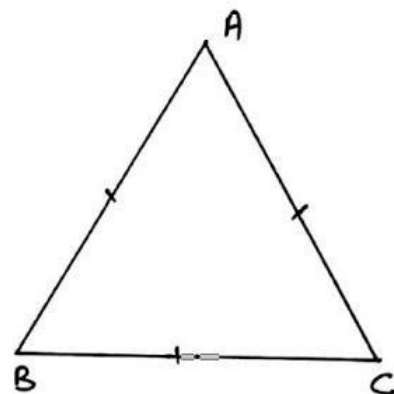
$$\angle ABC + \angle ACB + \angle BAC = 180^\circ \text{ [Angle sum property]}$$

$$\Rightarrow \angle ABC + \angle ABC + \angle ABC = 180^\circ \text{ [From (i) \& (ii)]}$$

$$\Rightarrow 3\angle ABC = 180^\circ$$

$$\Rightarrow \angle ABC = \frac{180^\circ}{3} = 60^\circ$$

Hence each angle of an equilateral triangle is 60°



Exercise 7.3

Q1 $\triangle ABC$ and $\triangle DBC$ are two isosceles triangles on the same base BC and vertices A and D are on the same side of BC .

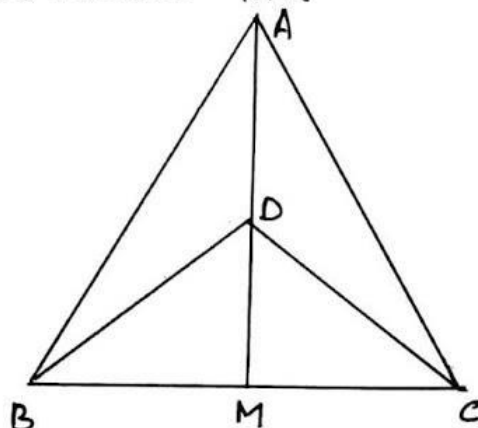
If AD is extended to intersect BC at P , show that

i) $\triangle ABD \cong \triangle ACD$

ii) $\triangle ABP \cong \triangle ACP$

iii) AP bisect $\angle A$ as well as $\angle D$

iv) AP is the perpendicular bisector of BC



Sol In $\triangle ABD$ and $\triangle ACD$, we have
 $AB = AC$ [Given]

$$BD = CD \quad [\text{Given}]$$

$$AD = AD \quad [\text{Common}]$$

$$\therefore \triangle ABD \cong \triangle ACD \quad [\text{SSS congruence}] \text{ proved } \text{---(i)}$$

ii) In $\triangle ABP$ and $\triangle ACP$, we have

$$AB = AC \quad [\text{Given}]$$

$$\angle BAP = \angle CAP \quad [\because \angle BAD = \angle CAD, \text{ by CPCT}] \text{---(ii)}$$

$$AP = AP \quad [\text{Common}]$$

$$\therefore \triangle ABP \cong \triangle ACP \quad [\text{SAS congruence}]$$

$$\text{iii) } \triangle ABD \cong \triangle ADC \quad [\text{From (i)}]$$

$$\Rightarrow \angle ADB = \angle ADC \quad [\text{CPCT}]$$

$$\Rightarrow 180^\circ - \angle ADB = 180^\circ - \angle ADC$$

$$\angle BDP = \angle CDP$$

$$\therefore \boxed{AP \text{ bisect } \angle D}$$

Also from equation (ii)

$$\angle BAP = \angle CAP$$

$$\therefore \boxed{AP \text{ bisect } \angle A}$$

iv) Now $BP = CP$

$$\text{and } \angle BPA = \angle CPA \quad [\text{By CPCT}]$$

$$\text{But } \angle BPA + \angle CPA = 180^\circ \quad [\text{Linear pair}]$$

$$\text{So } 2\angle BPA = 180^\circ$$

$$\text{or } \angle BPA = 90^\circ$$

Since $BP = CP$, therefore AP is perpendicular bisector of BC .

Hence Proved.

Q2 AD is an altitude of an isosceles triangle ABC in which $AB = AC$ Show that

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i) AD bisects BC

ii) AD bisects $\angle A$

Sol: $\rightarrow$ i) In $\triangle ABD$ and $\triangle ACD$, we have

$$\angle ADB = \angle ADC \text{ [Each } 90^\circ]$$

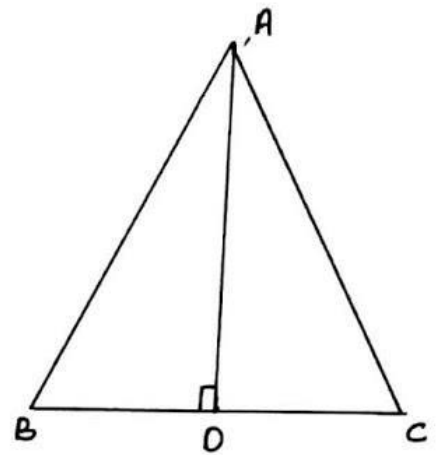
$$AB = AC \text{ [Given]}$$

$$AD = AD \text{ [Common]}$$

$$\therefore \triangle ABD \cong \triangle ACD \text{ [RHS Congruence]}$$

$$\therefore BD = CD \text{ [CPCT]}$$

Hence AD bisects BC



ii) Also $\angle BAD = \angle CAD$

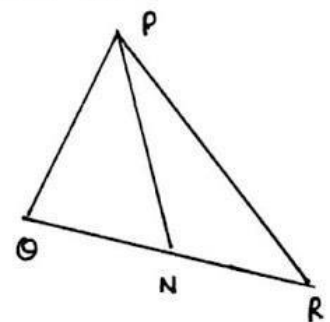
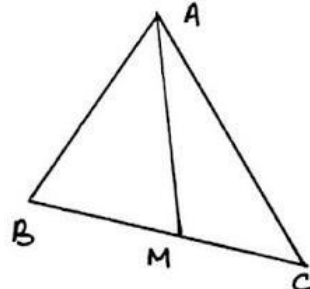
Hence AD bisects $\angle A$

Hence proved

Q3: $\rightarrow$ The sides AB and BC and median AM of one triangle ABC are respectively equal to sides PQ and QR and median PN of $\triangle PQR$ Show that

i) $\triangle ABM \cong \triangle PPN$

ii) $\triangle ABC \cong \triangle PQR$



Sol: $\rightarrow$ In $\triangle ABM$ and $\triangle PPN$ we have

$$BM = PN \text{ } [\because BC = QR]$$

$$\Rightarrow \frac{1}{2} BC = \frac{1}{2} QR$$

$$AB = PQ \text{ [Given]}$$

$$AM = PN \text{ [Given]}$$

$$\therefore \triangle ABM \cong \triangle PPN \text{ [SSS congruence]}$$

$$\Rightarrow \angle ABM = \angle PON \text{ [CPCT]}$$

Proved

ii) Now, in $\triangle ABC$ and $\triangle POR$, we have

$$AB = PO \text{ [Given]}$$

$$\angle ABC = \angle POR \text{ [Proved above]}$$

$$BC = OR \text{ [Given]}$$

$$\therefore \triangle ABC \cong \triangle POR \text{ [SAS congruence]}$$

Proved

Q4: $\rightarrow$ BE and CF are two equal altitudes of a triangle ABC. Using R.H.S Congruence rule, prove that the triangle ABC is isosceles.

Sol: $\rightarrow$ BE and CF are altitudes of a $\triangle ABC$

$$\therefore \angle BEC = \angle CFB = 90^\circ$$

Now, in right $\triangle BEC$ and $\triangle CFB$

$$BC = BC \text{ [Common]}$$

$$BE = CF \text{ [Given]}$$

$$\therefore \triangle BEC \cong \triangle CFB \text{ [By R.H.S Congruence rule]}$$

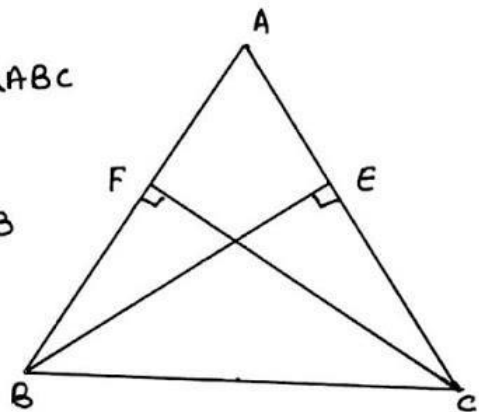
$$\therefore \angle BCE = \angle CBF \text{ [CPCT]}$$

$$\text{Now, in } \triangle ABC, \angle C = \angle B$$

$$\therefore AB = AC \text{ [side opposite to equal angles are equal]}$$

Hence, $\triangle ABC$ is an isosceles triangle

Hence proved.



Q5: $\rightarrow$ ABC is an isosceles triangle with $AB = AC$.

Draw $AP \perp BC$ to show that

$$\angle B = \angle C$$

Sol: $\rightarrow$ Draw $AP \perp BC$

In $\triangle ABP$ and $\triangle ACP$. We have

$$AB = AC \quad [\text{Given}]$$

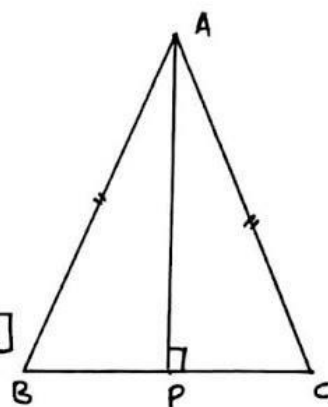
$$\angle APB = \angle APC \quad [\text{Each } 90^\circ]$$

$$AP = AP \quad [\text{Common}]$$

$$\therefore \triangle ABP \cong \triangle ACP \quad [\text{By RHS Congruence rule}]$$

$$\text{Also } \angle B = \angle C$$

Hence Proved [CPCT]



Exercise 7.4

Q1. Show that in a right angled triangle, the hypotenuse is the longest side.

Sol: $\rightarrow$ ABC is a right triangle, right angled at B

$$\text{Now } \angle A + \angle C = 90^\circ$$

$\Rightarrow$ Angles A and C are each less than 90°

$$\text{Now } \angle B > \angle A$$

$$\Rightarrow AC > BC \quad \text{---(i)}$$

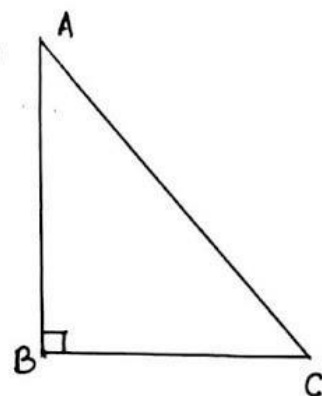
[Side opposite to greater angle is longer]

$$\text{Again } \angle B > \angle C$$

$$AC > AB \quad \text{---(ii)}$$

[Side opp to greater angle is longer]

Hence from (i) and (ii) we can say that AC (Hypotenuse) is the longest side



Q2: → In the figure side AB and AC of $\triangle ABC$ are extended to points P and Q respectively. Also $\angle PBC < \angle QCB$. Show that $AC > AB$

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Sol: → $\angle ABC + \angle PBC = 180^\circ$ [Linear Pair]

$$\Rightarrow \angle ABC = 180 - \angle PBC \text{ --- (i)}$$

$$\text{Similarly } \angle ACB = 180 - \angle QCB \text{ --- (ii)}$$

It is given that $\angle PBC < \angle QCB$

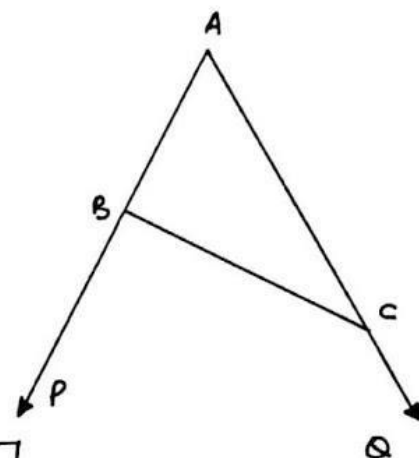
$$\therefore 180 - \angle QCB < 180 - \angle PBC$$

$$\Rightarrow \angle ACB < \angle ABC \text{ [From (i) and (ii)]}$$

$$\Rightarrow AB < AC$$

$$\text{or } \boxed{AC > AB}$$

Hence proved.



Q3: → In the figure $\angle B < \angle A$ and $\angle C < \angle D$. Show that $AD < BC$

Sol: →

$$\angle B < \angle A \text{ [Given]}$$

$$\therefore BO > AO \text{ --- (i)}$$

[Side opp to greater angle is longer]

$$\text{Also } \angle C < \angle D$$

$$\therefore CO > DO \text{ --- (ii)}$$

[Same reason]

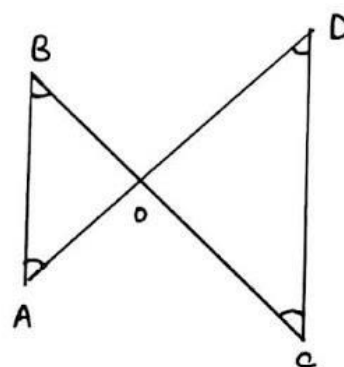
Adding (i) + (ii) we get

$$BO + CO > AO + DO$$

$$BC > AD$$

$$AD < BC$$

Hence Proved.



Q4 AB and CD are respectively the smallest and longest side of a quadrilateral ABCD. Show that $\angle A > \angle C$ and $\angle B > \angle D$

Sol : $\rightarrow$ Join AC

Mark the angles as shown in fig.

In $\triangle ABC$

$BC > AB$ [AB is the shortest side]

$$\Rightarrow \angle 2 > \angle 4 \quad \text{--- (i)}$$

[Angle opp to longer side is greater]

In $\triangle ADC$

$CD > AD$ [CD is the longest side]

$$\angle 1 > \angle 3 \quad \text{--- (ii)}$$

[Angle opposite to longer side is greater]

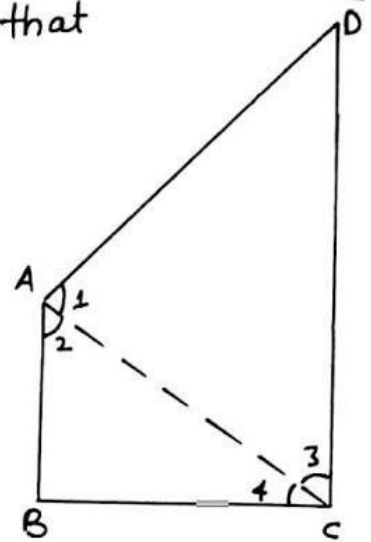
Adding (i) and (ii) we have

$$\angle 2 + \angle 1 > \angle 4 + \angle 3$$

$$\boxed{\angle A > \angle C}$$

Similarly by Joining BD, we can prove that

$$\boxed{\angle B > \angle D}$$



Q5 : $\rightarrow$ In the figure $PR > PO$ and PS bisects $\angle OPR$. Prove that $\angle PSR > \angle PSO$

Sol : $\rightarrow PR > PO$

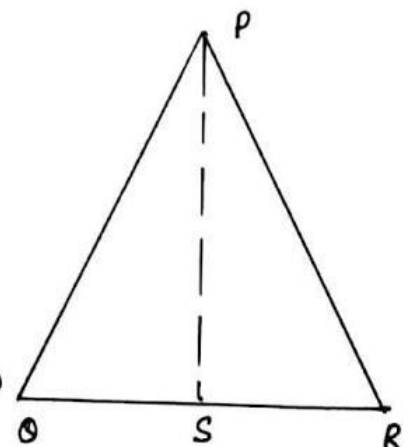
$$\angle POR > \angle PRO \quad \text{--- (i)}$$

[Angles opp to longer side is greater]

$$\angle OPS > \angle RPS \quad [\because PS \text{ bisects } \angle OPR] \quad \text{--- (ii)}$$

$$\text{In } \triangle POS, \angle POS + \angle OPS + \angle PSO = 180^\circ$$

$$\Rightarrow \angle PSO = 180^\circ - (\angle POS + \angle OPS) \quad \text{--- (iii)}$$



Similarly in $\triangle PSR$, $\angle PSR = 180 - (\angle PRS + \angle OPS)$ Page No. 16
 $\Rightarrow \angle PSR = 180^\circ - (\angle PRS + \angle OPS)$ — IV [from (ii)]

From (i)

we know that $\angle POS < \angle PSR$

So from (iii) and (iv), $\angle POS < \angle PSR$

$$\Rightarrow \angle PSR > \angle POS$$

Hence Proved

Q6 : $\rightarrow$ Show that of all the segments drawn from a given point not on it the perpendicular line segment is the shortest.

Sol : $\rightarrow$ We have A line $\vec{l}$ and O is a point not on $\vec{l}$

$$OP \perp \vec{l}$$

We have to prove that $OP < OO$, $OP < OR$ and

$$OP < OS$$

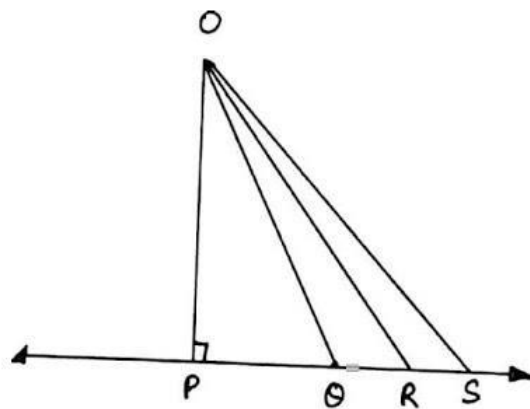
In $\triangle OPO$

$$\angle P = 90^\circ$$

$\therefore \angle O$ is an acute angle
 i.e. $\angle O < 90^\circ$

$$\therefore \angle O < \angle P$$

$$\text{Hence } OP < OO$$



[Side opposite to greater angle is longer]

Similarly, we can prove that OP is the Shorter than OR, OS etc.

Hence Proved